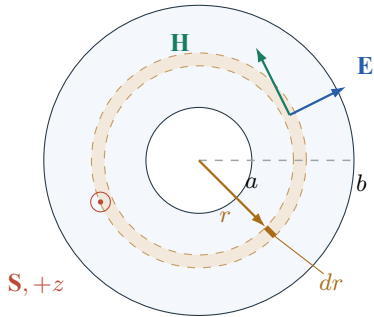
M7-049　同轴几何与特性阻抗



环带面积 $2\pi r\,dr$

$$L' = \frac{\mu_0}{2\pi}\ln\frac{b}{a}, \quad C' = \frac{2\pi\varepsilon_0}{\ln(b/a)}$$

$$Z_0 = 60\ln(2.30)\,\Omega \simeq 50\,\Omega$$

真空同轴线；忽略导体和介质损耗。